

Mini Test 5 Pathway 2

Pythagoras' Theorem

Show **ALL** working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Time Allocated 45 minutes

Resources Permitted- Calculators and 1 A4 Page of notes.

Name: Sol Ootious

Class: Master

Time: 45 minutes

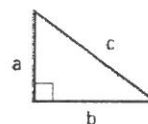
Total Marks: 20

1. Circle the correct formula for Pythagoras' Theorem for the triangle below:

a. $a^2 = c^2 + b^2$

b. $c^2 = a^2 + b^2$

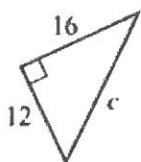
c. $b^2 = c^2 + b^2$



[1]

2. Use Pythagoras' Theorem to find the value of the pronumeral.

a.

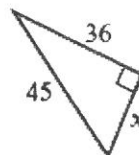


$$c^2 = 16^2 + 12^2 \checkmark$$

$$c = \sqrt{400}$$

$$\underline{c = 20} \checkmark$$

b.



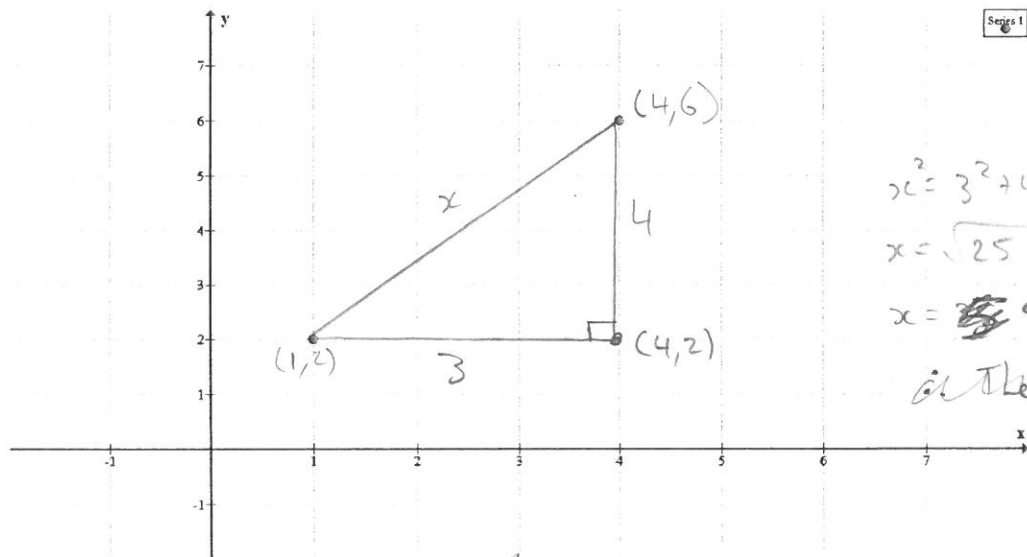
$$x^2 = 45^2 - 36^2 \checkmark$$

$$x = \sqrt{729}$$

$$\underline{x = 27} \checkmark$$

[4]

3. For each of the following use Pythagoras' Theorem to find the distance between the two points.
a.



$$x^2 = 3^2 + 4^2 \checkmark$$

$$x = \sqrt{25}$$

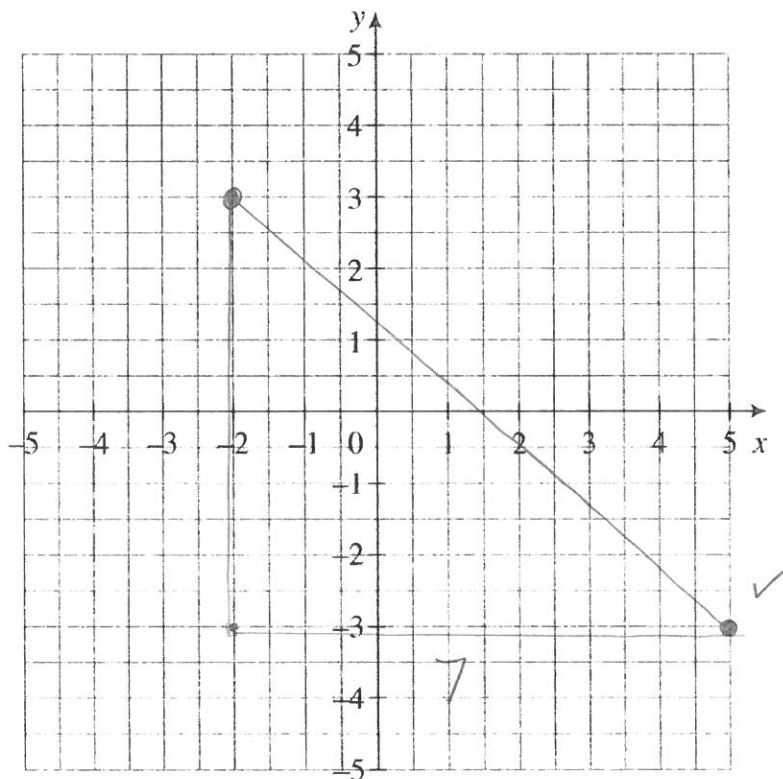
$$x = \cancel{4} 5$$

at the

The distance between the two points [2 marks]

is ~~4~~ 5 ✓

- b. Plot the points $(-2, 3)$ and $(5, -3)$, then find the length of the line correct to 3 dp.



$$\text{distance} = \sqrt{7^2 + 6^2}$$

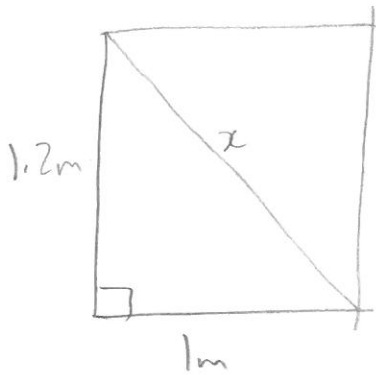
$$= \sqrt{85}$$

$$= \underline{9.220} \checkmark$$

[2 marks]

4. A gate is strengthened by adding a strut along each of its diagonals. If the gate is 1m wide and 1.2m high, find the length of each diagonal strut.

[2]

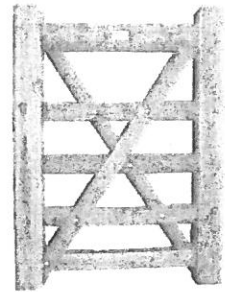


$$x^2 = 1.2^2 + 1^2 \quad \checkmark$$

$$x = \sqrt{2.44} \quad \checkmark$$

$$x = 1.56 \text{ m (2dp)} \quad \checkmark$$

$\therefore$ The length of each strut is 1.56m



5. I have a can of Coke Zero which is 15cm tall. My straw is 17cm long and fits diagonally inside the can. What must the radius of the can be?

[4]



$$d^2 = 17^2 - 15^2 \quad \checkmark$$

$$d = \sqrt{64} \quad \checkmark$$

$$d = 8 \text{ cm} \quad \checkmark$$

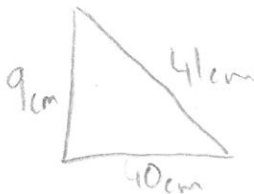
$$r = 4 \text{ cm} \quad \checkmark$$

$\therefore$ The radius of the can is 4cm



6. A triangle has side lengths of 9cm, 40cm and 41cm. Is it a right angled triangle? Give reasons for your answer

[2]



To be a right angled triangle

$$41^2 = 9^2 + 40^2$$

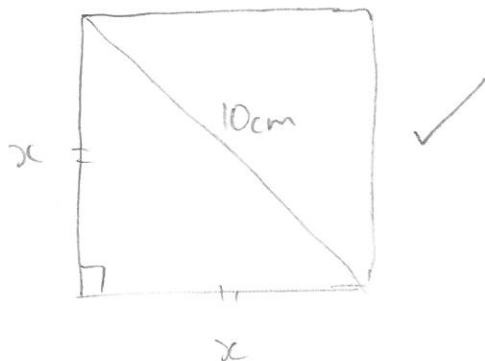
$$1681 = 81 + 1600 \quad \checkmark$$

$$1681 = 1681$$

$\therefore$ Yes, it is right angled due to pythagoras' theorem

7. A square has a diagonal of 10cm. What are the lengths of the sides? Round your answer to 2 decimal places.

[3]



$$10^2 = x^2 + x^2 \quad \checkmark$$

$$100 = 2x^2$$

$$50 = x^2$$

$$7.07 = x$$

$\therefore$ the side lengths of the square are 7.07 cm $\checkmark$

END OF TEST